



Elisabeth Uhlemann

Introducing Connected Vehicles

WELCOME NEW COLUMN EDITOR

With our first issue of 2015, *IEEE Vehicular Technology Magazine (VTM)* starts a new section on connected vehicles under the leadership of our new senior editor, Prof. Elisabeth Uhlemann from Mälardalen University in Sweden. Connected vehicles represent the intersection of our three fields of interest in the IEEE Vehicular Technology Society (VTS): mobile radio, automotive electronics, and transportation systems. To foster activities and the VTS presence in this area, the VTS decided to launch an IEEE Connected Vehicle Initiative in 2015. The creation of a new senior editor position is part of this initiative. With our new column, *VTM* will regularly inform VTS members of developments in this industry as the presence of wireless communications in the automotive and transportation industries is becoming more ubiquitous and we prepare for the launch of vehicle-to-vehicle communications and vehicle-to-infrastructure communications technologies from 2016 to 2017. In addition, *VTM* will promote the publication of technical articles in this area.

Prof. Uhlemann was previously the associate editor in charge of the networked vehicular area within the mobile radio

section. She has been involved in several European projects studying communication requirements for traffic-safety applications in vehicular networks and actively collaborates with the Volvo Group. She also contributed to the European Intelligent Transportation System (ITS) communications architecture, and she actively participates as a technical expert in standardization within the European Telecommunications Standards Institute Technical Committee on ITS. She is also a very active VTS member. She is the vice chair of the Swedish VT/COM/IT Chapter and the founder and organizer of the VTS Workshop on Wireless Vehicular Communications. We are very pleased to welcome her on board, and we are sure our members will enjoy her regular columns on connected vehicles.

We will keep you informed of the new activities launched within the IEEE VTS Connected Vehicle Initiative. If you would like to get involved, do not hesitate to contact Javier Gozalvez, who will lead the initiative, at j.gozalvez@umh.es.

— Javier Gozalvez

The term *connected vehicles* refers to applications, services, and technologies that connect a vehicle to its surroundings. Adopting a definition similar to that of AUTO Connected Car News, a connected vehicle is basically the presence of devices in a vehicle that connect to other devices within the same vehicle and/or devices, networks, applications, and services outside the vehicle. Applications

include everything from traffic safety and efficiency, infotainment, parking assistance, roadside assistance, remote diagnostics, and telematics to autonomous self-driving vehicles and global positioning systems (GPS). Typically, vehicles that include interactive advanced driver-assistance systems (ADASs) and cooperative intelligent transport systems (C-ITS) can be regarded as connected. Connected-vehicle safety applications are designed to increase situation awareness and mitigate traffic accidents

through vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications. ADAS technology can be based on vision/camera systems, sensor technology, vehicle data networks, V2V, or V2I systems. Features may include adaptive cruise control, automate braking, incorporate GPS and traffic warnings, connect to smartphones, alert the driver to hazards, and keep the driver aware of what is in the blind spot. V2V communication technology could mitigate traffic collisions and improve traffic

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CONNECTED-VEHICLE SAFETY APPLICATIONS ARE DESIGNED TO INCREASE SITUATION AWARENESS AND MITIGATE TRAFFIC ACCIDENTS THROUGH V2V AND V2I COMMUNICATIONS.

congestion by exchanging basic safety information such as location, speed, and direction between vehicles within range of each other. It can supplement active safety features, such as forward collision warning and blind-spot detection.

This column will begin with an overview of some important enabling technologies including C-ITS, its standards, and acronyms before summarizing some news and recent developments within the area of connected vehicles.

Cooperative Intelligent Transport Systems

C-ITS, where vehicles cooperate by exchanging messages wirelessly to avoid, e.g., hazardous road traffic situations are currently receiving a great deal of attention around the world. There are intense research activities and standardization efforts ongoing within this area. Further, major field operational tests are planned and underway. Depending on the application, there are mainly two wireless technologies considered for C-ITS: the short-range communication technology IEEE 802.11p and cellular networks such as third-generation (3G) or long-term evolution (LTE). The latter depends on a centralized network topology where all data traffic passes through a base station (BS). IEEE 802.11p, on the other hand, does not rely on access points but relies on direct communication between ITS stations by forming a vehicular ad hoc network (VANET). Although no access point is present in 802.11p, there can still be fixed ITS stations (e.g., ITS-equipped traffic lights) offering services to mobile ITS stations (e.g., vehicles), and the distinction between a mobile and a fixed ITS station is made at higher layers in the protocol stack.

C-ITS applications can roughly be categorized into road traffic safety, road traffic efficiency, and value-added services. Road traffic safety applications have stringent requirements for both bounded delay and high reliability, concurrently. Therefore, protocol stacks dedicated to supporting this type of application have been developed in the United States and Europe. The common denominators for these stacks are the communication technology 802.11p and simple network/transport protocols with low overhead, allowing them to pass through the protocol faster. Road traffic efficiency applications and value-added services can use 802.11p technology and/or 3G/LTE together with the network protocol [Internet protocol version 6 (IPv6)] supporting Internet connections. Value-added services may be, e.g., announcements of commercial services such as advertisements for hotels, petrol stations, grocery stores, and so on. Examples of road traffic safety applications are warnings of lane changes, emergency vehicles approaching, stationary vehicles, road conditions, and road work. Efficiency applications aim at enhancing the road traffic flow and reducing carbon dioxide emissions and pollution, through, e.g., green-light optimal speed advisory, traffic-light optimization, in-vehicle signage, and enhanced route guidance. However, the border between road traffic safety and road traffic efficiency applications is not clear cut. If an accident occurs, e.g., a vehicle suddenly breaks down in the middle of the street (stationary vehicle), the enhanced route guidance application can advise drivers to take an alternative route before they reach the stationary vehicle and end up in a queue with no

possibility to turn around. In other words, the safety application can, in certain situations, trigger efficiency applications and vice versa.

Different C-ITS applications have different communication requirements, and, thus, 3G/LTE and IEEE 802.11p are not two competing technologies; rather, they complement each other. One of the main advantages of 802.11p is that it does not rely on coverage by access points or BSs to disseminate information locally around the ITS stations. Information can also be transmitted in a certain direction on the highway using geographical routing (georouting). In addition, by using the ad hoc communication mode of IEEE 802.11p, delays can be kept lower since no detour around a BS is necessary. C-ITS applications that do not have short delay requirements or rely on information being spread regionally rather than locally can utilize 3G/LTE. For example, a fixed ITS station using 802.11p can act as an information provider to services offered by 3G/LTE by transmitting service announcements to surrounding vehicles, which, in turn, can connect to the 3G/LTE network themselves to retrieve more information.

Standards and Acronyms

The standardization of vehicular communications took off in 1999 when the U.S. Federal Communications Commission (FCC) allocated a 75-MHz band at 5.850–5.925 GHz especially intended for ITS: “to improve traveler safety, decrease traffic congestion, facilitate the reduction of air pollution, and help to conserve vital fossil fuels.” The American Society for Testing and Materials (ASTM) was commissioned to bring forward a standard, and it did so by starting from the wireless local area network standard IEEE 802.11 and making some minor adjustments to better fit the high-speed vehicular environment. The ASTM E2213-03 standard was approved in 2003 and contained three layers: application, data link, and physical.

IEEE Wireless Access in Vehicular Environment

In 2003, the IEEE took over the work from the ASTM, and in November 2004, the 802.11p task group was formed. The task group extended the protocol stack to include layers to support Internet access, and this more holistic view has been termed *wireless access in vehicular environment* (WAVE). The WAVE protocol stack includes the physical, data link, network, and transport layers and is divided into two parts: safety applications and nonsafety applications. The nonsafety applications use the traditional Internet protocol stack containing IPv6, the user datagram protocol for connectionless services, as well as the transmission control protocol for connection-oriented services. The safety applications use one common protocol for the network and transport layers called the *WAVE short-message protocol*, standardized through IEEE 1609.3 and with security from IEEE 1609.2. Both parts of the protocol stack share the same data link layer and physical layer for transmission. The data link layer uses IEEE 1609.4 for multichannel operation according to the FCC frequency regulations. The MAC sublayer and the physical layer are derived from IEEE 802.11-2012. In 2010, the vehicular “profile” of 802.11 was approved and termed *802.11p*. It has been incorporated in the latest version of IEEE 802.11-2012, and 802.11p is now classified as superseded.

IEEE 802.11p

IEEE 802.11-2012 contains two basic network topologies: the infrastructure basic service set (BSS) and the independent BSS (IBSS). The former contains an access point, and data traffic usually takes a detour through the access point even when two stations are closely located. The IBSS is a set of stations directly communicating with each other, which is also known as an ad hoc network. Both of these topologies are intended for nomadic devices, and synchronization is required

between stations via beacons. Furthermore, they are identified with a unique BSS identification (BSSID). Association and authentication are required in infrastructure BSS (containing access points), whereas in IBSS (ad hoc mode), association is not used and communication can take place in an unauthenticated mode.

In 802.11p mode, authentication, association, and security between stations are disabled at the MAC sublayer. This is done since the BSS authentication and association procedures are time consuming, and in a VANET where stations are highly mobile, this transaction may not be completed until the stations are out of radio range from one another. However, this also implies that active scanning and passive scanning of the BSS and IBSS are disabled. The scanning on frequency channels for the station to join an existing network is no longer enabled. Therefore, the implementation of 802.11p in the vehicular environment requires predetermined frequency channels to be set in the management. To distinguish between communication within a BSS and outside of it, the network identification (BSSID) is set to a wildcard in every frame transmitted in an 802.11p network.

IEEE 802.11-2012 offers several physical layers and one common MAC sublayer with quality of service (QoS) support. The MAC algorithm deployed by 802.11p is found in the IEEE 802.11-2012, and it is called *enhanced distributed coordination function*. It is based on the basic distributed coordination function (DCF) but adds QoS attributes through the former amendment, IEEE 802.11e. The DCF is a carrier sense multiple-access (CSMA) with collision-avoidance algorithm, and DCF was already present in the first version of IEEE 802.11 released in 1997. IEEE 802.11p uses the orthogonal frequency-division multiplexing physical layer detailed in Clause 18 of 802.11 (also known as IEEE 802.11a), and it has support for QoS

through IEEE 802.11e (approved in 2004 and enrolled in the base standard in 2007).

DSRC Versus DSRC

The term dedicated short-range communication (DSRC) is sometimes used to denote IEEE 802.11p or the ad hoc part of WAVE, especially in the United States or Australia. However, an electronic toll collection (ETC) system called DSRC had already been used in Europe and Japan since 2003. The ETC is a master-slave system operating on the 5.8-GHz band, using a fixed roadside unit (RSU) and short-range communications. In Europe, it is standardized by the European Committee for Standardization (CEN) as DSRC. Since the introduction of WAVE, the term *DSRC* most likely refers to the IEEE 802.11p-based VANET system operating at 5.9 GHz, whereas the master-slave ETC system operating at the 5.8-GHz band is referred to as *CEN-DSRC*.

International Organization for Standardization Communications Access for Land Mobiles

The International Organization for Standardization developed a framework termed *communications access for land mobiles* (CALM). The idea with CALM is to use all types of already existing wireless access technologies, such as 2G/3G/LTE, wireless broadband access (e.g., WiMAX), Wi-Fi, infrared, and IEEE 802.11p, to provide seamless wireless connection to the end users and applications. In the protocol stack above the different wireless carriers (i.e., the network layer), the IPv6 is found, joining all different access technologies. The idea is to develop applications and services related to connected vehicles completely independent of the underlying wireless access technology. For supporting VANET, CALM M5 has been developed, which is based on IEEE 802.11p. However, not all ITS applications can use IPv6 due to, e.g., the rapidly changing environment.

Therefore, a low-overhead network and transport layer protocol has been developed called *CALM—non-IP networking*.

European Telecommunications Standards Institute Technical Committee on Intelligent Transport Systems

In 2007, the European Telecommunications Standards Institute (ETSI) established a technical committee (TC) on ITS, which is responsible for bringing standards forward in Europe for all parts of the protocol stack. In 2009, the European Commission (EC) issued mandate M/453, “Standardisation Mandate Addressed to CEN, CENELEC, and ETSI in the Field of Information and Communication Technologies to Support the Interoperability of Cooperative Systems for Intelligent Transport in the European community.” The ETSI and CEN responded to the mandate, which is legally binding. The ETSI is responsible for developing the whole protocol stack, including vehicle-centric road traffic safety applications, whereas applications oriented toward road traffic efficiency utilizing road infrastructure are the responsibility of the CEN.

In Europe, 30 MHz has been set aside for vehicular communications at 5.875–5.905 GHz, solely intended for road traffic safety applications. Nonsafety-related applications are directed to a 20-MHz band at 5.855–5.875 GHz. The dedicated frequency bands have been divided into 10-MHz frequency channels. Due to the proximity of these bands to the frequency band used for ETC in Europe (5.795–5.805 GHz), the ETSI TC ITS must also develop mitigation techniques to avoid interfering with the CEN-DSRC systems. There is no cost associated with using this frequency band (it is license-free). However, the usage of it is regulated in EN 302 571 specifying requirements on output power limits, spectrum masks, and so on.

A basic set of applications has been defined by the ETSI TC ITS,

which has been grouped into road safety, traffic efficiency, and other applications. The European protocol stack includes a facilities layer situated in between the network/transport layer and the applications. The session, presentation, and application layers have been merged into the facilities layer, which handles functions like the local dynamic map (LDM). The LDM part belonging to the vehicle is to be standardized within the ETSI TC ITS, whereas the LDM utilized by RSUs will be developed within the CEN.

In February 2014, the CEN and ETSI delivered the first set of standards for C-ITS. The so-called Release 1 specifications will enable vehicles made by different manufacturers to communicate with each other and with the road infrastructure systems.

Distributed Congestion Control

The first access-layer technology standardized within the ETSI TC ITS is termed *ITS-G5*, and it is based on IEEE 802.11p, with the additional requirement on distributed congestion control (DCC). DCC is a collection of methods for controlling the network load to avoid unstable behavior. The MAC method used by IEEE 802.11-2012 is CSMA. This MAC method has trouble accommodating all broadcast transmissions in ad hoc mode when the density of stations increases. Simultaneous transmissions are then more likely to occur by geographically close or colocated stations (within radio range of each other). DCC is a way of mitigating these problems. It is based on the idea that each and every station monitors the channel activity, and, whenever some predetermined threshold on the channel occupancy is exceeded, countermeasures are taken to avoid unstable behavior. There are three main tools available for DCC: transmit rate control (TRC), transmit power control (TPC), and transmit data rate control (TDC). TRC implies that the number of generated packets is reduced when the channel load is high. TPC controls the output

power on a per-packet basis and decreases and increases the output power depending on channel activity. The last tool, TDC, is possible in 802.11 since it offers several transfer rates and implies that when the instantaneous channel load increases, stations can use higher transfer rates to decrease the channel occupancy for every packet, i.e., the packet duration is decreased. The DCC approach within the ETSI TC ITS is a cross-layer issue that has been divided between the management plane, access layer, network/transport layer, and facilities layer. In TS 102 687, a DCC framework is detailed, and this TS is also a requirement in the access layer specification EN 302 663.

GeoNetworking

The ETSI TC ITS has also developed a GeoNetworking concept at the network/transport layers, where four different communication scenarios are foreseen: point-to-point, point-to-multipoint, GeoAnycast, and GeoBroadcast. Geographical addressing and geographical forwarding are two cornerstones in GeoNetworking. The addresses used for forwarding packets among the stations are based on the geographical positions of the stations. The forwarding itself relies on the fact that every station has a perception of its part of the network, in other words, the nearest neighbor of the station and its position. The station does not maintain routing tables but instead it keeps a list of neighbors it can hear (receive packets from), and based on the geographical address in an incoming packet, the station forwards the packet if suitable. With GeoNetworking, packets can be addressed to certain geographical regions of interest without knowing if there are stations in the destined area or not. The two central GeoNetworking standards in the network layer are divided into 1) media-dependent functionalities and 2) media-independent functionalities. The former is specified in TS 102 636-4-1 and deals with the DCC mechanism on the network layer

specifically tailored toward the access technology ITS-G5. The media-independent functionalities are found in TS 102 636-4-2 and specify packet types that can be used in the different GeoNetworking scenarios. It is also possible to transmit IPv6 datagrams over the GeoNetworking protocols, and this is described in TS 102 636-6-1.

Recent News on Connected Vehicles

eCall: Automatic Emergency Call System in All New European Cars by 2018

The in-vehicle eCall system uses 112 emergency call technology to alert emergency services to serious road accidents automatically. This enables them to decide immediately on the type and size of rescue operation needed, which, in turn, helps them to arrive faster, save lives, reduce the severity of injuries, and cut the cost of traffic jams.

Some manufacturers are already offering eCall-type services to drivers through private call centers. The deal provides for the coexistence of two systems, public eCall and eCall-supported third-party services (TPSS), provided that 112-based eCall is always automatically available if TPSS fail to work and that vehicle owners may choose public eCall services rather than private ones at any time.

As a public service, eCall will be free of charge for all citizens, whatever car they drive and whatever its purchase price. The new rules will ensure that eCall works only as a safety device. It will be illegal to use it to track a driver's movements or to misuse location data, which must be sent only to emergency services.

Each year, emergency services across the European Union deal with road accidents; in 2013, accidents took 26,000 lives. The members of the European Parliament (MEPs) strengthened the draft law's data-protection clause to preclude tracking of an eCall-equipped vehicle before the accident occurs. Under the agreed deal, the

automatic call would give the emergency only basic, minimum data such as the class of vehicle, the type of fuel used, the time of the accident, and the exact location. MEPs also amended the draft law to ensure that data gathered by emergency centers or their service partners must not be transferred to third parties without explicit consent of the person concerned, the "data subject." Manufacturers will also have to ensure that the eCall technology design permits full and permanent deletion of the data gathered. Clear information about the processing of eCall data would have to be included in the car owner's manual and available online, the MEPs added.

All new models of passenger cars and light commercial vehicles will have to be equipped with the eCall system no later than 31 March 2018. In the following three years, the EC will assess whether eCall should be extended to other vehicles, such as buses, coaches, or trucks, says the agreement text.

Cadillac Adds Streaming Video to Rearview Mirror to Enhance Driver Vision and Safety

Cadillac will add high-resolution streaming video to the function of a traditional rearview mirror. The streaming video mirror improves field of vision by an estimated 300%, roughly four times greater than a standard rearview mirror. In addition to the increased field of view, the technology eliminates any rear seat, rear pillar, or passenger obstructions, allowing the driver an unimpeded view of the lanes behind the vehicle and traditional blind spots.

Connected-Vehicle Technologies on Select Cadillac 2017 Models

General Motors (GM) announced that Cadillac will begin offering advanced "intelligent and connected" vehicle technologies on certain 2017 model

year vehicles. Super Cruise, the working name for GM's automated driving technology, will offer customers a new type of driving experience that includes hands-off lane following, braking, and speed control in certain highway driving conditions. The system is designed to increase the comfort of an attentive driver on freeways, both in bumper-to-bumper traffic and on long road trips.

V2X Chip Set for Connected Vehicles in Mass Production

NXP Semiconductors announced that V2X chip sets will be put into high-volume manufacturing. NXP will supply Delphi Automotive PLC with its RoadLINK chip set for V2V and V2I communication. Delphi's platform is expected to be on the roads in as little as two years. Using NXP chips with V2X hardware security module combined with application software from Cohda Wireless, Delphi's platform allows alerts to be delivered to vehicles from other cars and surrounding infrastructure, such as traffic lights and signage to alert drivers about potentially hazardous traffic situations even beyond the line of sight, complementing ADASs like radar.

Predictive Traffic Forecasts for Better Estimated Arrival Times

HERE, a Nokia company focusing on navigation, mapping, and location experiences, has launched Predictive Traffic, a new traffic forecasting product that can anticipate future traffic conditions in real time. HERE Predictive Traffic considers over 100 different traffic behavior profiles per road segment to provide up-to-the-minute traffic forecasts as far as 12 h into the future. HERE devised a proprietary mechanism that selects the appropriate profile from this exhaustive collection and combines it

(continued on page 31)

Klaus Moessner is a professor in mobile communications in the Institute for Communication Systems at the University of Surrey, United Kingdom. He earned the Dipl-Ing (FH) from The University of Applied Sciences in Offenburg, Germany; the M.Sc. from Brunel University, United Kingdom; and the Ph.D. from the University of Surrey, United Kingdom.

His research interests include reconfigurability on different levels, including object and function

virtualization, management, and control of virtualized resources, and cognitive networks. The interests map to the crowd sensing, activity and situation recognition, and cognitive systems areas, and the smart city and smart transport domains. He is involved in the investigation and teaching of mobile operating systems, mobile service platforms, service-oriented architectures, mobile service delivery, and service enablers.

He has been involved in different roles in many U.K.- as well as European community-supported research projects in the area of the Internet of Things. This includes the projects EyeHUB, M:Ciudad, OUTSMART, iCore, Probe-IT, IoT.est, COSMOS, SocloTal, iKAAS, and FrontierCities.

He leads the work area on system architecture and coexistence in the Fifth-Generation Innovation Centre at the University of Surrey. **VT**

TRANSPORTATION SYSTEMS *(continued from page 22)*

The PKP Pendolino has seven cars and can carry up to 402 people in three classes: first, second, and dedicated closed compartments for families. The cars are all equipped with air conditioning, passenger information on light-emitting diode screens, table and electrical sockets for each passenger, high luggage capacity, and bicycle storage.

InterCity Express (EIC) Premium is the most comfortable train category that PKP Intercity offers its passengers. It is also a whole new dimension and standard of traveling in Poland because PKP's ultimate goal is the convenience and safety of the passenger. Premium EIC trains connect north and south Poland, operating between Tri-City, Warsaw, Krakow, Katowice, and Wroclaw.

Tickets are available over the counter, from ticket machines, on the Web site, and through mobile applications. They cannot be purchased on the train from the conductor.

First-class passengers receive a meal and two drinks of their choice, while second-class passengers receive a snack and a choice of one drink.

The new trains have been adapted to the needs of people who are blind or visually impaired. For example, the seats are marked in Braille. For people with reduced mobility, a ramp supported by the staff of the train has been supplied.

The commissioning of the new Pendolino trains follows a contract worth €665 million awarded by PKP Intercity in 2011 to supply 20 high-speed trains, the full maintenance of the fleet for up

to 17 years, and the construction of a maintenance depot in Warsaw.

In September 2014, the Polish Office of Rail Transport, responsible for the supervision of the railway sector's safety in Poland, certified the Pendolino trains. During the tests, the Pendolino beat its own speed record with a speed of 182 mi/h, establishing Poland's highest ever speed record on rail.

The European Union (EU) is supporting sustainable transportation initiatives and allocated €74 million for this high-speed project in Poland. An additional €10 billion will be granted within the EU cohesion fund 2014–2020 to further support the development of the Polish railway network. With over 2,500 mi of track, Poland has the third biggest railway network in Europe. **VT**

CONNECTED VEHICLES *(continued from page 27)*

with real-time conditions to provide travelers with the best travel time estimates for their destination. Drivers using Predictive Traffic saw estimated arrival times that were up to 20% more accurate for journeys over 30 min in length.

Automakers, transportation agencies, and other companies can build applications that seamlessly integrate HERE predictive traffic data across all screens. Preplanned trips could be synced to the cloud, e.g., allowing for automatic integration with the car and

real-time notifications via connected devices. Dynamic message signs on highways could broadcast better travel times to the public. The product will initially be available to original equipment manufacturers in the United States, Canada, and Germany. **VT**